PROGRAM CODE :

```
clc; clear; close all;

% Parameters Am = 1; fm = 100; Ac = 5; fc = 1000; fs = 20*fc; kp = pi/2;

% Time Vector t = 0:1/fs:4/fm;

% Message Signal m = Amsin(2pifmt);

% PM Signal pm = Accos(2pifct + kp*m);

% Phase Deviation beta = kp*Am;

% FFT N = length(pm); f = (-N/2:N/2-1)*(fs/N); PM_FFT = abs(fftshift(fft(pm)))/N;

% Power Spectral Density PSD = (abs(fftshift(fft(pm))).^2)/(N*fs);

% Bandwidth BW = 2*(beta+1)*fm;

fprintf('Phase Deviation = %.2f radians\n',beta); fprintf('Estimated Bandwidth = %.2f Hz\n',BW);

% Plotting figure;

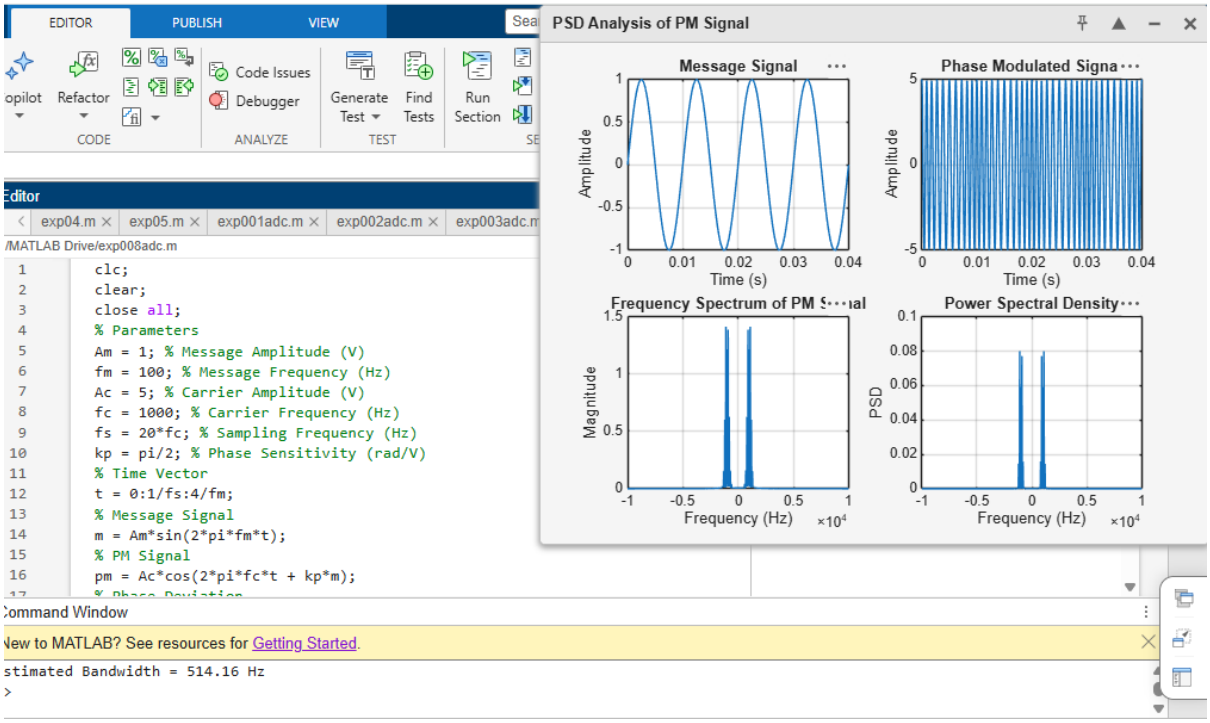
subplot(2,2,1); plot(t,m); title('Message Signal'); xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(2,2,2); plot(t,pm); title('Phase Modulated Signal'); xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(2,2,3); plot(f,PM_FFT); title('Frequency Spectrum of PM Signal'); xlabel('Frequency (Hz)'); ylabel('Magnitude'); grid on;

subplot(2,2,4); plot(f,PSD); title('Power Spectral Density'); xlabel('Frequency (Hz)'); ylabel('PSD'); grid on;
```

OUTPUT:


EDITOR
PUBLISH
VIEW
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
CODE
ANALYZE
TEST
exp04.m
exp05.m
exp001adc.m
exp002adc.m
/MATLAB Drive/exp008adc.m
clc;
clear;
close all;
% Parameters
Am = 1; % Message Amplitude (V)
fm = 100; % Message Frequency (Hz)
Ac = 5; % Carrier Amplitude (V)
fc = 1000; % Carrier Frequency (Hz)
fs = 20*fc; % Sampling Frequency (Hz)
kp = pi/2; % Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;
% Message Signal
m = Am*sin(2*pi*fm*t);
% PM Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
Command Window
New to MATLAB? See resources for Getting Started.
stimated Bandwidth = 514.16 Hz
PSD Analysis of PM Signal
Message Signal
Phase Modulated Signa
Amplitude
Time (s)
Frequency Spectrum of PM S
Power Spectral Density
Magnitude
PSD
Frequency (Hz)